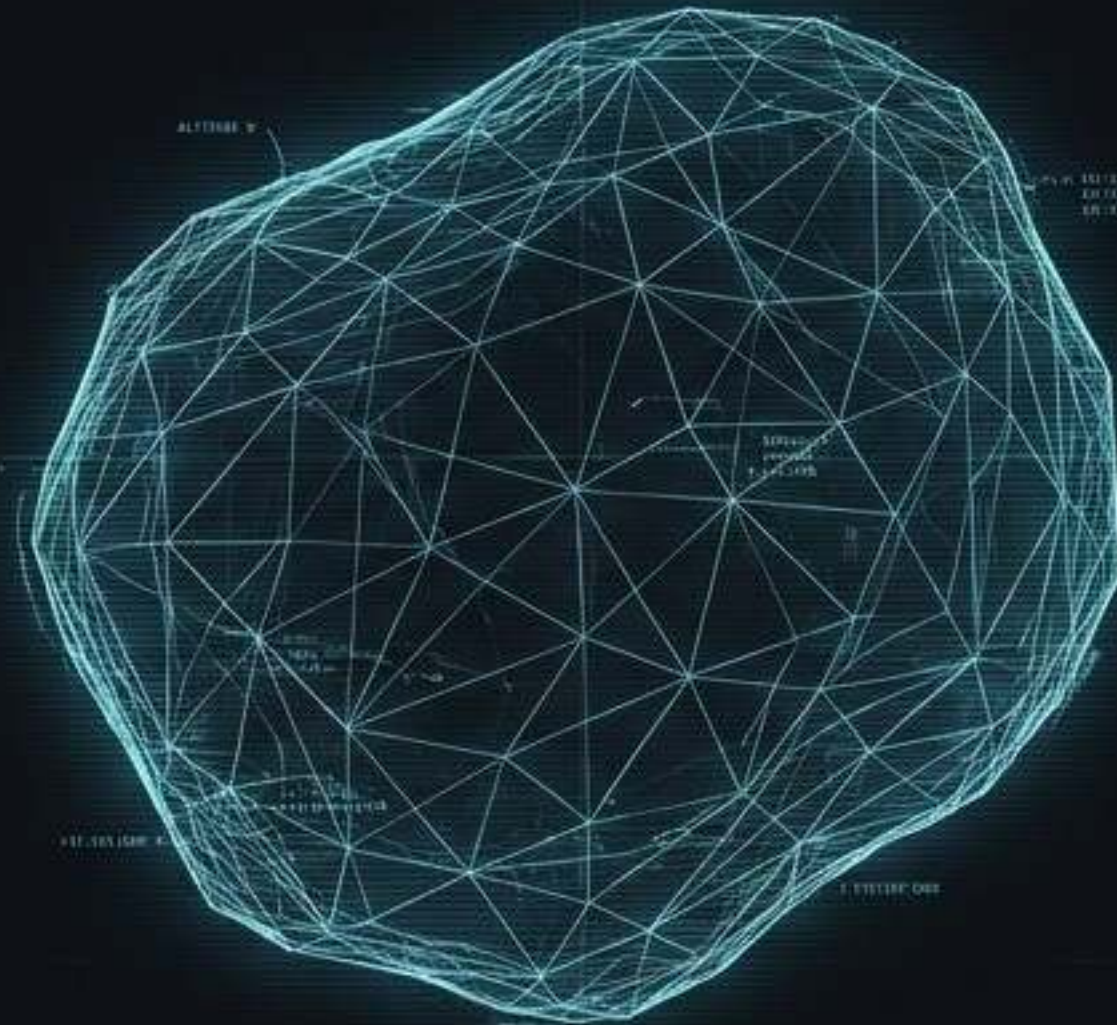


ORBITAL TRACKING ACTIVE
SYSTEM STATUS: NOMINAL

DATA STREAM: SECURE
CONNECTION: STABLE

PLANETARY DEFENSE

AI-DRIVEN ASTEROID CLASSIFICATION



PROJECT SCOPE: BIG DATA ANALYTICS | A.A. 2025 - 2026

DATA SOURCE: NASA NEO (NEAR-EARTH OBJECT) DATASET

AUTHORS: FABRIZIO M. CRIMI 593588 | MICHELE FALCONETTI 596924 | MICHELE STORELLI 598652

MISSION OBJECTIVE: PREDICT HAZARDOUS IMPACTORS VIA DEEP LEARNING

MISSION CLOCK: T-MEMO 55:56:50
LOCATION: DEEP SPACE NETWORK

POWER LEVEL: 100%
INITIALIZING: PDI_CORE_V0.1

MISSION OBJECTIVES AND CONTEXT

Binary Classification



OBJECTIVE

Develop predictive models to classify Near-Earth Objects (NEOs) as Hazardous or Non-Hazardous.

METHODOLOGY

Performance comparison between Logistic Regression (Baseline) and Artificial Neural Networks (Deep Learning).

DATA SOURCE

NASA Open Data (JPL/CNEOS).

INTELLIGENCE SOURCE: DATASET

SOURCE METADATA

Origin: NASA Official Databases via Kaggle

Period: 1910–2024

Volume: 90,000+ Records

Format: CSV (Structured)

NASA - Nearest Earth Objects

A cumulative data for Nearest Earth Objects by NASA



id	name	# est_diameter...	# est_diameter...	# relative_velo...	# miss_distance	orbiting_body	sentry_object	# absolute_ma...	hazardous
2162635	162235 (2000 SST64)	1.1982700007	2.6794149958	13569.2492241012	54839744.08294605	Earth	False	16.73	False
2277475	277475 (2005 WK4)	0.26S8	0.5594486694	73583.7286894981	61438126.32333093	Earth	False	20.0	True
2512244	S12244 (2015 YE18)	0.7220295677	1.6145071127	114258.2521990512	49798724.94445679	Earth	False	17.83	False
3396000	(2012 BV13)	0.096555147	0.2577948049	24764.3031380016	25444577.72075825	Earth	False	22.2	False
3667127	(2014 GEYS)	0.2530088879	0.3702167609	42797.3337887224	46575557.00130072	Earth	False	20.09	True
54188966	(2021 GYY3)	0.0665543223	0.0812093344	34297.5877883029	40555661.22792288	Earth	False	24.32	False
54189957	(2021 PY40) AF2)	0.1116148941	3.8753743344	27592.5787789963	20065921.41664897	Earth	False	20.95	False
54230078	(2021 XD6) AF2)	0.0033278866	0.0119135161	57544.4700827552	55115019.25807114	Earth	False	23.49	False

DATA HANDLING AND CLEANING



Missing Values: 0
Intact Dataset

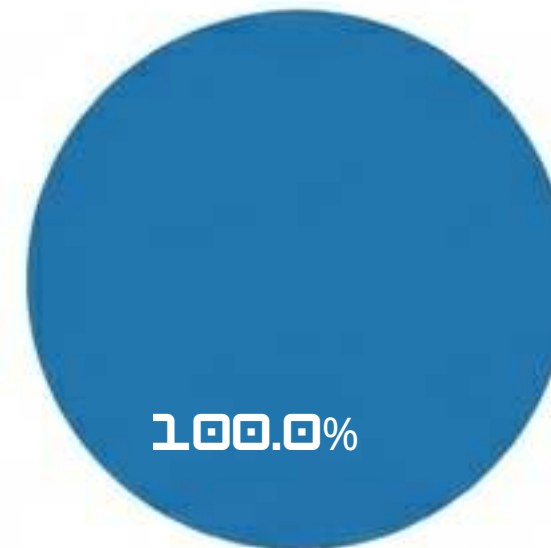


Noise removal:
ID, Name



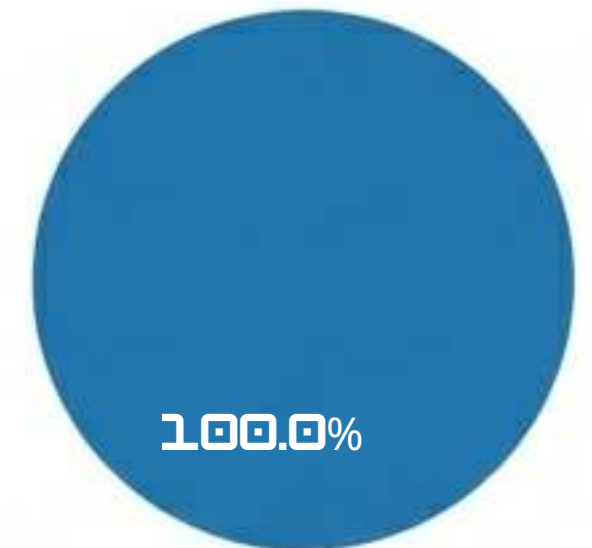
Zero Variance Check

Distribution of orbiting_body Distribution of sentry_object



100.0%

Earth

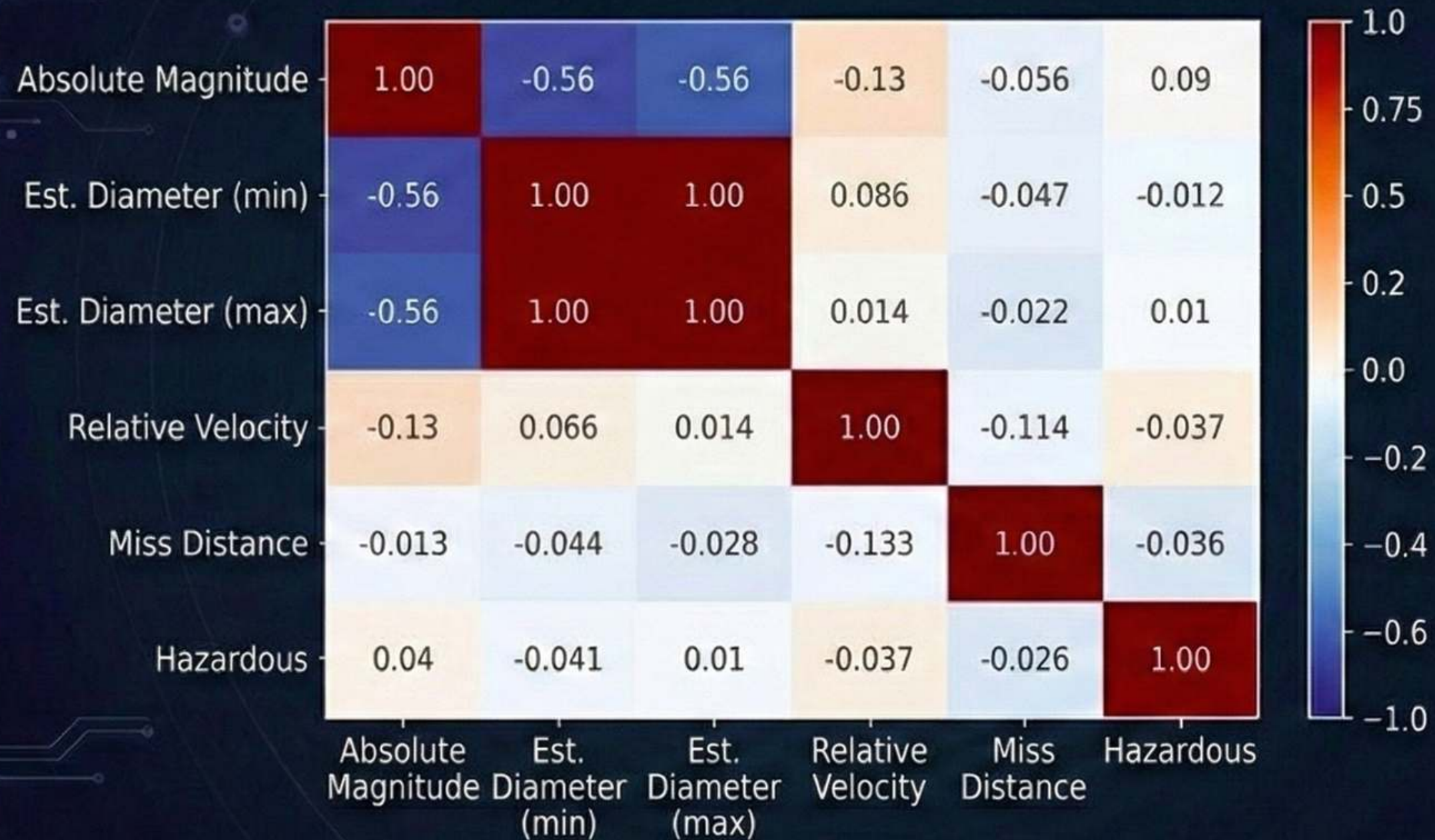


100.0%

False

CLEANING PROTOCOL: Constant variables provide zero information gain.
Removed to reduce dimensionality

EXPLORATORY ANALYSIS: CORRELATIONS



Data Redundancy:
Strong collinearity
between diameter
estimates.

Magnitude:
inversely
proportional to
size .

SYSTEM CALIBRATION: Z-SCORE STANDARDIZATION

BEFORE

Raw Distribution



Velocity: 57633.98
Diameter: 0.052

SCALING ALGORITHM
StandardScaler
(Mean=0, Std=1)

AFTER

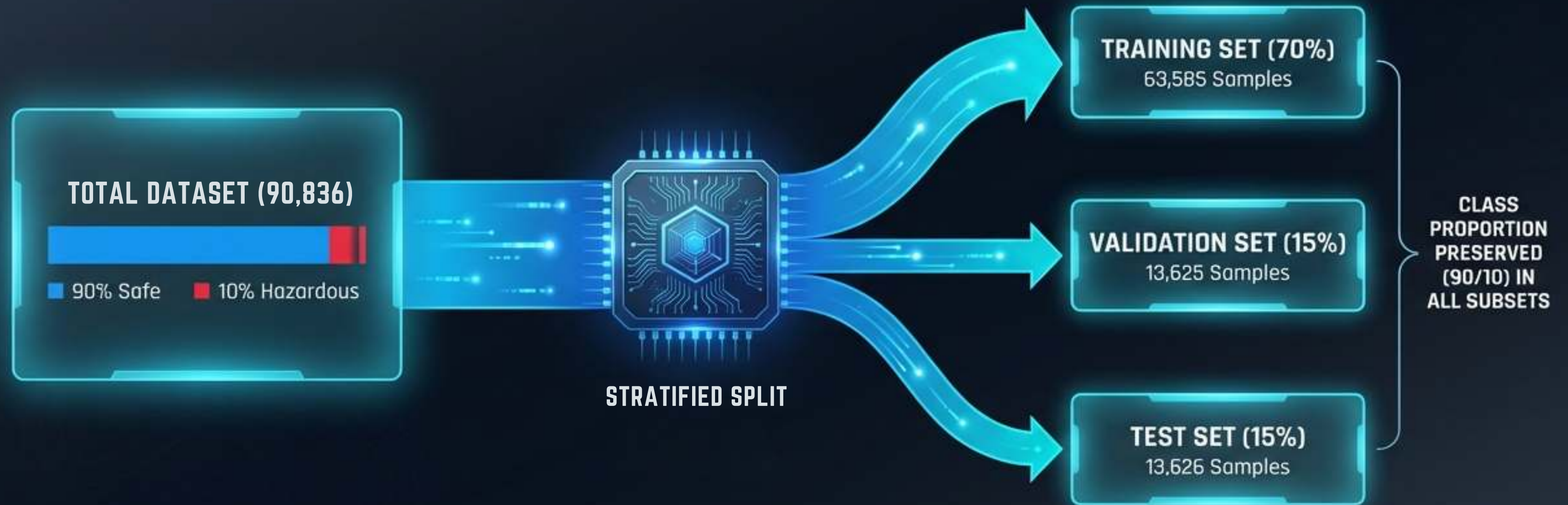
Normalized Input



est_diameter_min	relative_velocity	miss_distance
2.884250	-0.866651	-0.948186
-0.609815	1.419350	-1.547573
-0.846423	0.424010	-1.220878

Why? Essential for Gradient Descent convergence and Neural Network stability.

STRATEGIC PARTITIONING: STRATIFIED SAMPLING



Why? Ensures representation of minority classes, preventing bias in training and testing phases.

THE METRIC STRATEGY

RECALL (PRIORITY)

SYSTEM STATUS:
CALIBRATING



Metric purpose: The ability to capture all hazardous objects.

Cost of False Negative:
IMPACT EVENT.

MCC SCORE

TIME ELAPSED:
00:01:45



Matthews Correlation Coefficient.

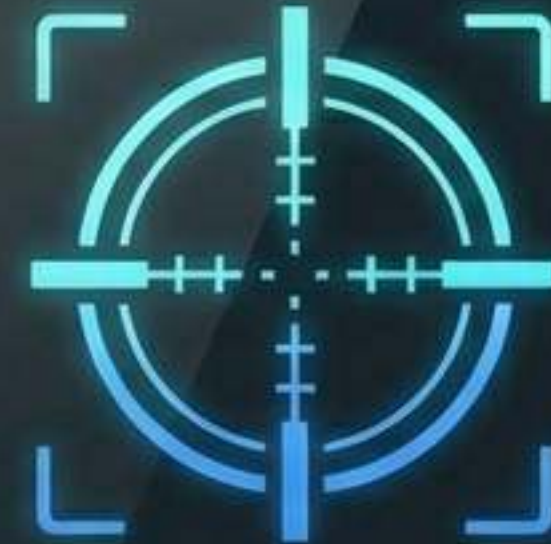
- Handles class imbalance better than F1.
- Range: -1 to +1.

-1.00  +1.00

PRECISION

TIME ELAPSED:
00:01:45

DATA STREAM:
STABLE

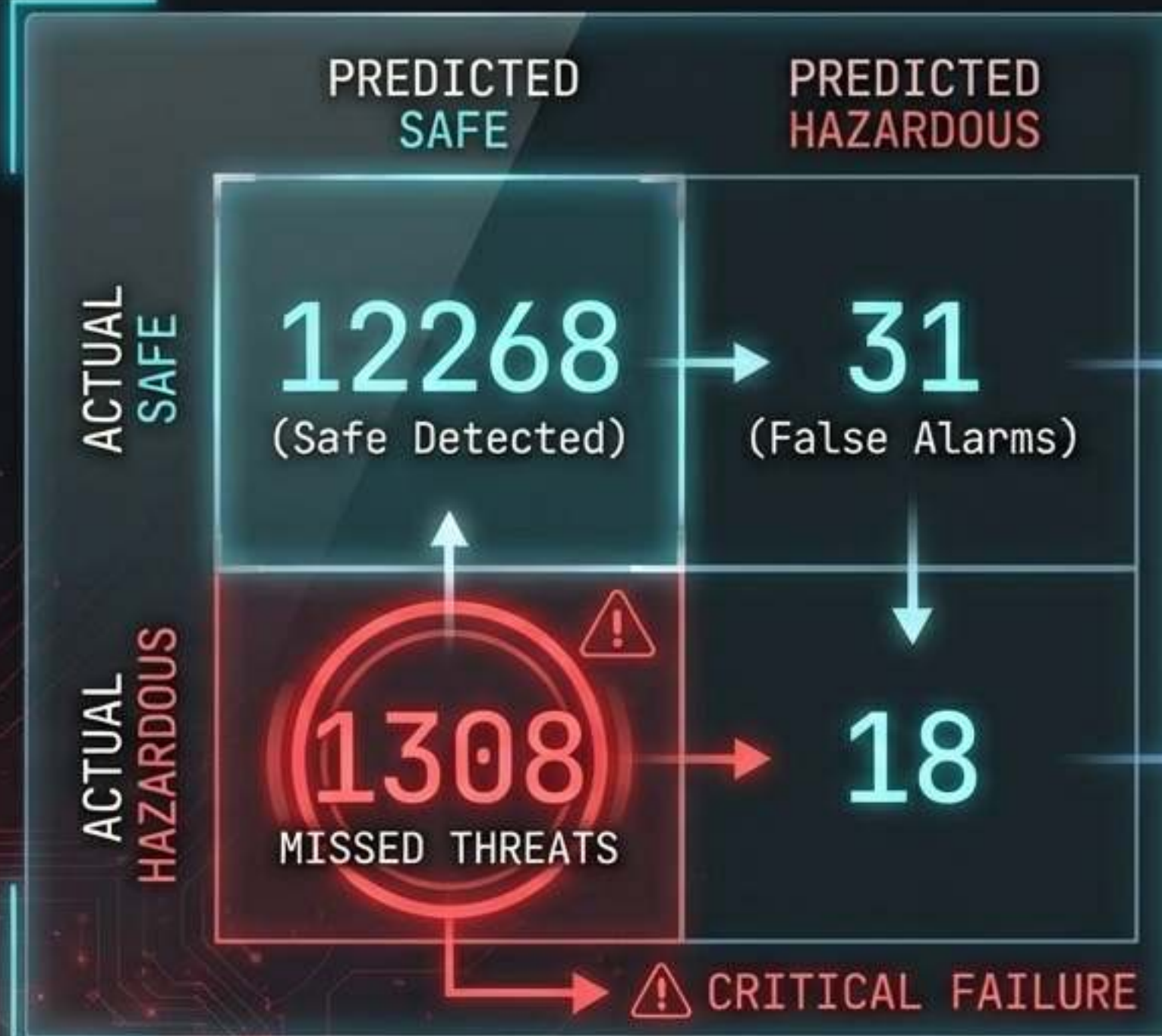


Secondary Priority.

Cost of False Positive:
Wasted telescope resources (Acceptable).

BASELINE SIMULATION: LOGISTIC REGRESSION

UNWEIGHTED MODEL RUN



PERFORMANCE METRICS

ACCURACY: ~90%



RECALL (HAZARDOUS): 0.01 !



VERDICT:

CRITICAL FAILURE.

The model ignores threats to maximize 'Safe' accuracy.

CORRECTIVE MEASURE: CLASS WEIGHTING

LOGISTIC REGRESSION WEIGHTED RUN

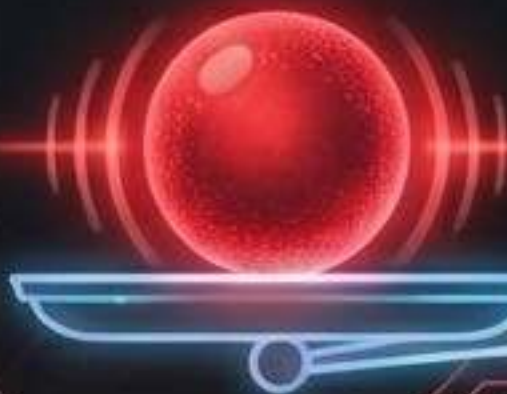
SAFE

Weight: LOW



HAZARDOUS

Weight: HIGH



`class_weight='balanced'`

The model is now punished heavily for missing a Hazardous object.

RESULTS

RECALL: 0.01 → 0.93 ↑

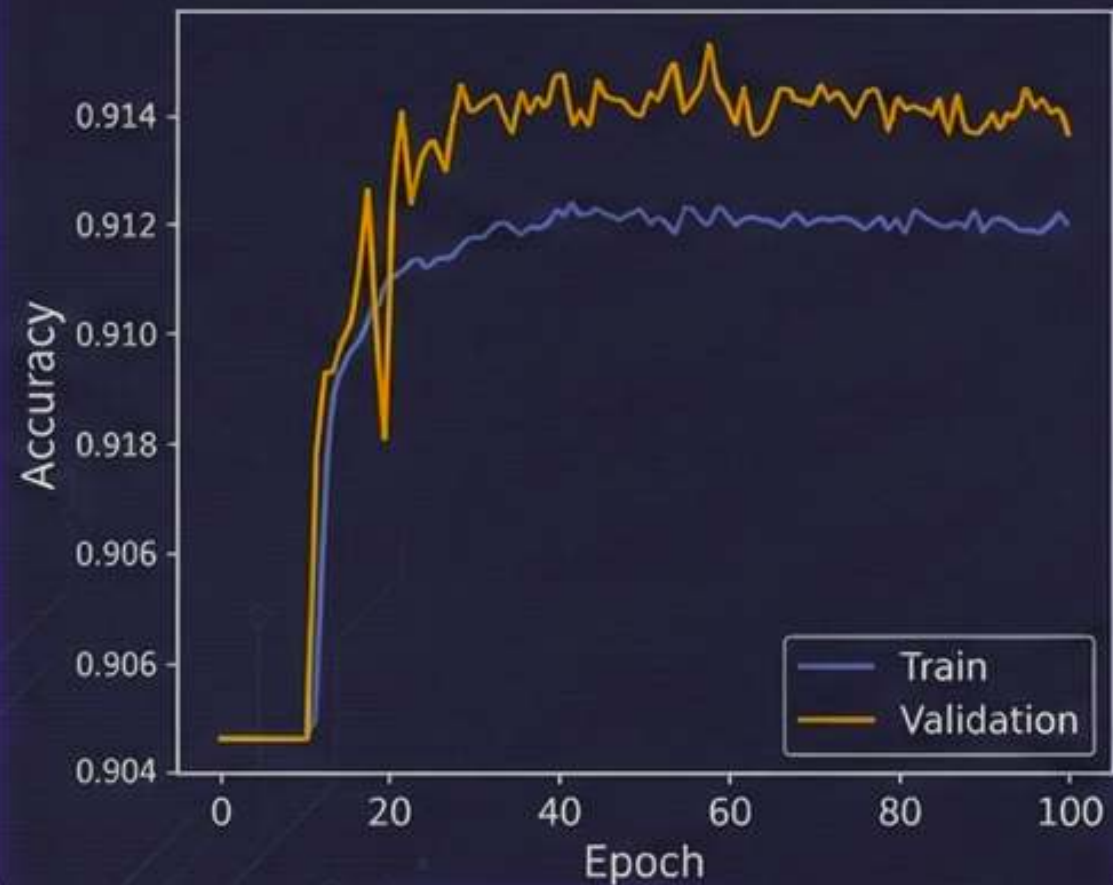
PRECISION:
Drops to 0.31 ↓

STATUS:
THREATS DETECTED.
HIGH FALSE ALARM RATE.

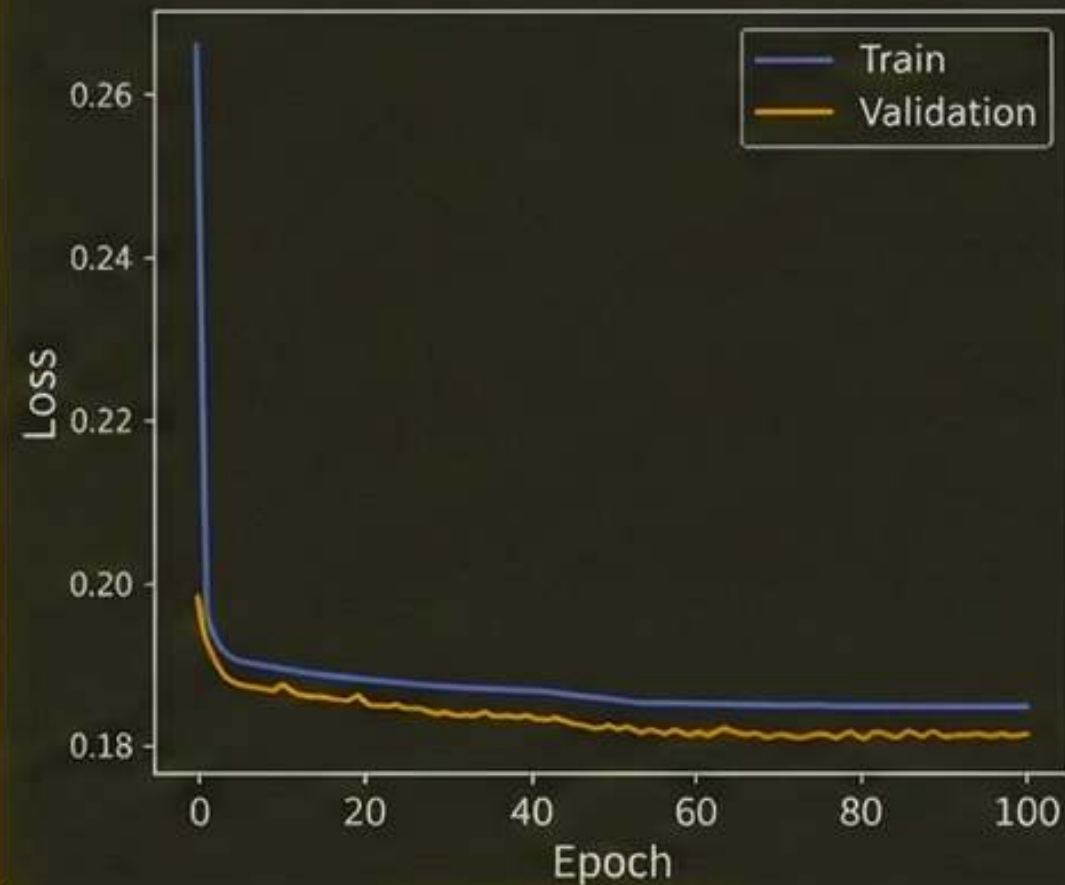
NEURAL NETWORK: RUN 1 (UNWEIGHTED)

1 HIDDEN LAYER | ADAM OPTIMIZER

Model Accuracy

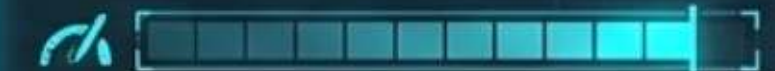


Model Loss

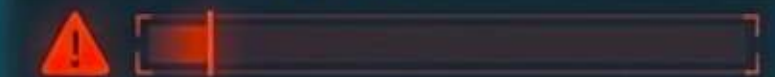


PERFORMANCE METRICS

ACCURACY: 91%



RECALL: 0.14



LOSS FUNCTION:
BINARY CROSSENTROPY

VERDICT:

FAILURE. Deep Learning without class weighting mimics the naive baseline. It optimizes for the majority class.

DEEP LEARNING + CLASS WEIGHTS

RUN 2:1 HIDDEN LAYER | SUCCESSFUL CONFIGURATION



RECALL: 0.99 ↑

MCC: 0.477

MISSED THREATS: ~17 (out of 1326)

OPERATIONAL BREAKTHROUGH. By combining Neural Networks with Class Weighting, we recover nearly 100% of hazardous objects.

TIME ELAPSED: 75:12:45

DATA STREAM: **STABLE**

ARCHITECTURE VARIATION: 2 HIDDEN LAYERS

TESTING COMPLEXITY



RECALL: 0.98
(Slight Decrease)

MCC: 0.482
(Minimal Gain)

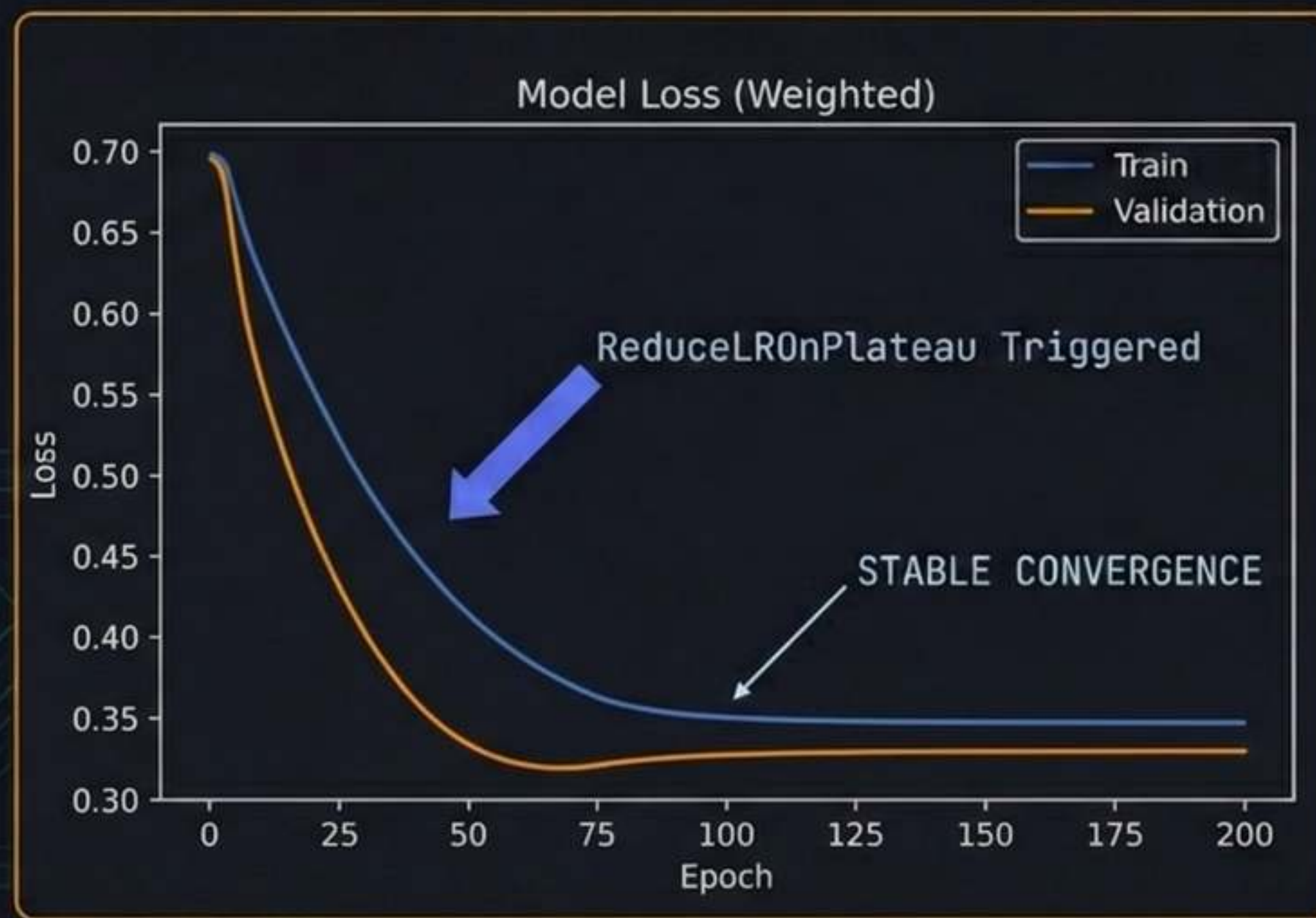
CONCLUSION: Added complexity reduced stability without meaningful performance gain.

TIME ELAPSED: 92:45:10

DATA STREAM: VOLATILE

OPTIMIZATION: LEARNING RATE SCHEDULER

FINE-TUNING CONVERGENCE



RECALL: 0.99 ↑

MCC: 0.467

OPERATIONAL EFFECTIVENESS: THE FACE-OFF

BASELINE (UNWEIGHTED LOGREG)



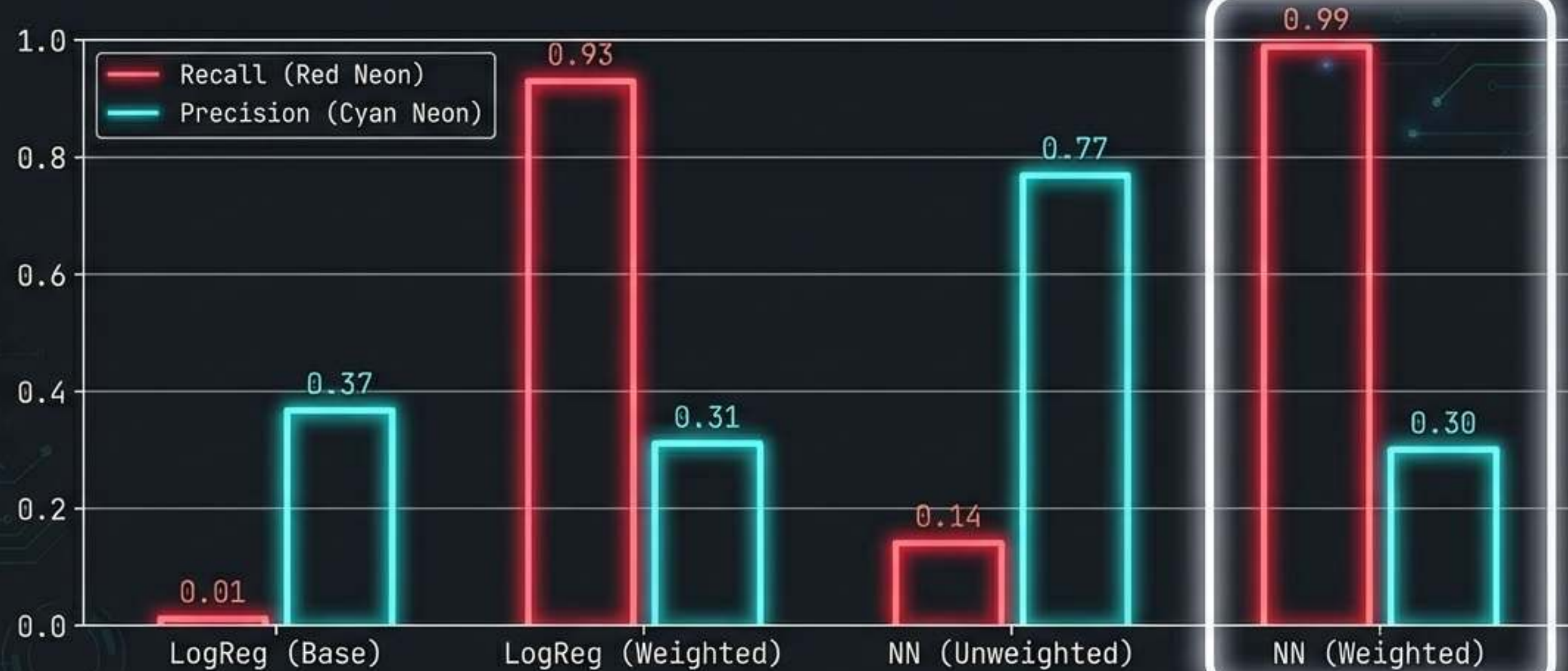
FINAL MODEL (WEIGHTED NN)





PERFORMANCE SHOWDOWN

MISSION READY



MODEL PERFORMANCE SUMMARY

Main metrics extracted from the tests performed on the various model. The F1-score column includes two values: the first refers specifically to the hazardous class of asteroids, while the second (in parentheses) indicates the overall weighted average.

Model	Accuracy	F1-score (Hazardous / Weighted)	MCC	Cohen's Kappa
Baseline Random (fifty-fifty)	50.59%	0.17 (0.60)	0.016	0.010
Baseline MAJ1 (95% Pred Hazardous)	13.68%	0.18 (0.10)	0.008	0.001
Baseline MAJ0 (95% Pred Safe)	86.18%	0.07 (0.84)	-0.000	-0.000
Logistic Regression (Real / Balanced Weights)	78.33%	0.46 (0.82)	0.452	0.362
Neural Network (Unweighted)	91.00%	0.24 (0.88)	0.3099	0.2190
Weighted Neural Network (1 Hidden Layer)	78.00%	0.47 (0.82)	0.4769	0.3750
Weighted Neural Network (2 Hidden Layers)	79.00%	0.48 (0.83)	0.4826	0.3842
Weighted Neural Network (Learning Rate Scheduler)	77.00%	0.46 (0.82)	0.4669	0.3621

Key takeaways emerging from the data:

As seen from the Baseline MAJ0 and the Unweighted Neural Network, having the highest overall Accuracy and weighted F1-score of the entire test set does not translate into a functional model: the F1-score on actual threats collapses, and the MCC uncovers the deception with zero or low scores. The Deep Learning models trained with Balanced Weights (Class Weighting) show a lower global Accuracy compared to the unweighted model, but they manage to maximize the MCC and Cohen's Kappa, demonstrating the effective achievement of the classification goal.

FUTURE TRAJECTORIES & SYSTEM EVOLUTION



OPTIMIZATION

Implement SMOTE (Synthetic Minority Over-sampling Technique) to generate synthetic hazard samples.



FEATURE SELECTION

Reduce input dimensionality by removing correlated diameter metrics.



EVOLUTION

Integration of Time-Series Analysis. Move from static snapshots to temporal orbital paths.

Machine Learning is effective, but requires targeted balancing for asymmetric risk scenarios.